Problem · Topic 3 · 4 points · 50 min

P12 — AGRODRON: the second year of a drone start-up that closed its first year with a loss

*The company **AGRODRON LEVANTE S.L.** was incorporated on 1 January 2017 (01/01/2017) by a group of agricultural engineers with a registered paid-in capital of 4.500 m.u., having an objective of providing precision crop-monitoring services by drone and distributing organic fertilizers in the region of Murcia.*

The following information applies to fiscal year 2017:

To finance the operations of the company, on 1 January 2017 (01/01/2017) a bank granted the company a loan of 1.500 m.u. at 6% annual interest, with a duration of 5 years, i.e. until 31 December 2021 (31/12/2021). Each 31 December, the company must return to the bank an equal principal amount of 300 m.u. and should also pay the due annual interest (6%) calculated from the loan value at the beginning of each year.

During 2017, the company invested in a fleet of agricultural drones worth 2.400 m.u. The useful life of the drones purchased was 8 years (amortized linearly, with no residual value) and at the end of 2017 the company recorded in accumulated depreciation an amount of 75 m.u. for those assets. Also, a further investment of 400 m.u. in mapping software licenses was made on 31 December 2017 (31/12/2017) to process aerial images. They use these licenses for 4 years (linear amortization with residual value zero). In addition, the company spent 800 m.u. to purchase a plot of rural land for its flight testing base. Most of 2017 was devoted to obtaining civil aviation permits and only the actual business work was started in the final months of the year. At 31 December 2017 (31/12/2017), its balance sheet reflected a loss of 220 m.u., there was no stock (stock in stock) nor pending customer or supplier balances, and the cash in hand and bank account was 1.955 m.u.

The following information applies to fiscal year 2018:

- *For aerial crop monitoring, the average number of subscribed farms during 2018 was 400 with an average revenue of 0,45 m.u. / month per client in subscription fees. Additionally, the company billed 840 m.u. for the direct sales of organic fertilizers. At the end of 2018, AGRODRON had accounts receivable of 310 m.u. from their customers.*
- *The company purchased supplies of organic fertilizers during 2018 which amounted to 500 m.u. Of these purchases, 70% was paid immediately in cash and the payment of the remaining 30% was deferred to the following year. Year 2018 sales consumed 80% of the purchased goods.*
- *AGRODRON operates from rented hangars that have a monthly cost (from 1 January 2018 (01/01/2018)) of 25 m.u. / month, paid in cash. Fixed salaries and Social Security costs (contribution of the company to the Social Security) of the permanent employees amounted to 60 m.u. per month (from 1 January 2018 (01/01/2018)) and were paid in full in 2018. To respond to increased demand during the harvest campaign, the company hired external pilots whose variable wages amounted in 2018 to 192 m.u. (paid in cash).*

- *Electricity, insurance, and other administrative expenditures totaled 180 m.u. in 2018. The company postponed the payment of 40% of this amount to the following year. Furthermore, on 1 July 2018 (01/07/2018), the company acquired thermal cameras for its drones for a value of 200 m.u., paying 50% in cash on that date and leaving the remainder payable to 9 months, i.e. on 1 April 2019 (01/04/2019). This equipment has a useful life of 5 years and will be amortized linearly with a residual value of 20%.*
- *The financial results of the company are declared at 23:59 hrs on 31 December 2018 (31/12/2018). The applicable corporation tax rate was 25% on the Profits obtained and is payable in the first quarter of 2019. In 2018, the results were positive: after offsetting in full the loss of the previous year, the company decided to distribute 30% of the remaining net profit after taxes as dividends (paid at 24:00 on 31 December 2018 (31/12/2018)) and retain the rest in the company.*

Required:

1. *Construct the Income Statement (Profit and Loss Account) of AGRODRON LEVANTE S.L. for the period ending 31 December 2018 (31/12/2018), and deduce the exact date in 2017 on which the fleet of agricultural drones was acquired.*
2. *Construct and justify the balance sheet at 24:00 on 31 December 2018 (31/12/2018), detailing all intermediate cash and accrual calculations.*
3. *Determine the values of the ROA and ROE ratios in 2018.*
4. *Indicate which items of the balance sheet at 31 December 2018 (31/12/2018) would be affected and the exact amount of the effect if the decision was made before the end of 2018 to distribute the same dividend, but to pay it on 30 June 2019 (30/06/2019), and assuming simultaneously that the company had decided not to offset the loss of 2017 before calculating the 30% dividend payout.*

Setup

Hypotheses:

1. The tax of 25% is applied to the whole profit before taxes of 2018; the loss of 2017 is offset afterwards, against the net profit, as the statement orders ("after offsetting in full the loss of the previous year, … distribute 30% of the remaining net profit after taxes").
2. The software licences were bought on the last day of 2017, so their amortization starts in 2018 (a full year).
3. The fertilizers not consumed (20% of the purchases) stay in stock at cost.

The balance at 31/12/2017, rebuilt from the data (needed to start 2018):

Assets	**m.u.**	**Liabilities and owners equity**	**m.u.**
Land	800	Paid-in capital	4.500
Drones (2.400 − 75)	2.325	Losses of 2017	(220)
Software licences	400	**Owners equity**	**4.280**
Cash	1.955	Bank loan due 2019–2021	900

Assets	m.u.	Liabilities and owners equity	m.u.
		Bank loan due 31/12/2018	300
Total assets	**5.480**	**Total liabilities and equity**	**5.480**

The loan was 1.500 and 300 were repaid on 31/12/2017, so 1.200 remain at the start of 2018.

Extension (not defined in the course decks). ROA = Income from operations/Total assets and ROE = Net income/Owners equity, on the balance at 24:00 on 31/12/2018.

Part (a) · 1.25 points

Prepare the 2018 Income Statement and deduce the purchase date of the drones

Step 1. Purchase date of the drones

$$D_{\text{drones}} = \frac{2.400 - 0}{8} = 300 \text{ m.u./year}, \qquad \frac{75}{300} \times 12 = 3 \text{ months} \;\Rightarrow\; \text{bought on 1 October 2017 (01/10/2017)}$$

Step 2. Revenue

$$\text{Subscriptions} = 400 \times 0{,}45 \times 12 = 2.160, \qquad \text{Revenue} = 2.160 + 840 = 3.000 \text{ m.u.}$$

Step 3. Fertilizers consumed

$$\text{Consumed} = 0{,}80 \times 500 = 400, \qquad \text{Stock at } 31/12/2018 = 500 - 400 = 100 \text{ m.u.}$$

Step 4. Depreciation and amortization

$$D_{\text{drones}} = 300, \qquad D_{\text{software}} = \frac{400}{4} = 100, \qquad D_{\text{cameras}} = \frac{200 - 0{,}20 \times 200}{5} \times \frac{6}{12} = 32 \times \frac{6}{12} = 16$$

$$D_{2018} = 300 + 100 + 16 = 416 \text{ m.u.}$$

Step 5. Interest

$$I_{2018} = 0{,}06 \times 1.200 = 72 \text{ m.u.} \quad \text{(loan value at the beginning of 2018)}$$

Step 6. Income Statement 2018

Line	Computation	m.u.
Subscription fees	$400 \times 0{,}45 \times 12$	2.160
Sales of fertilizers		840
Revenue		**3.000**
– Fertilizers consumed	Step 3	400
– Hangar rent	12×25	300
– Fixed salaries and Social Security	12×60	720
– Variable wages of external pilots		192
– Electricity, insurance and administration		180
– Depreciation and amortization	Step 4	416
= Income from operations	$3.000 - 2.208$	**792**
– Financial expenses (interest)	Step 5	72
= Profit before taxes	$792 - 72$	**720**
– Corporation tax (25%)	$0{,}25 \times 720$	180
= Net profit	$720 - 180$	**540**

KEY RESULTS

- Drones bought on 1 October 2017 (01/10/2017).
- Revenue 3.000; fertilizers consumed 400; depreciation and amortization 416 m.u.
- Income from operations 792; interest 72; profit before taxes 720; tax 180; net profit 540 m.u.

Part (b) · 1.5 points

Prepare the balance sheet at 24:00 on 31 December 2018

Step 1. Distribution

$$\text{Remaining profit} = 540 - 220 = 320, \qquad \text{Dividends} = 0{,}30 \times 320 = 96, \qquad \text{Reserves} = 320 - 96 = 224 \text{ m.u.}$$

Step 2. Collections and payments

- Customers: $3.000 - 310 = 2.690$ collected.
- Fertilizers: $0{,}70 \times 500 = 350$ paid, 150 owed.
- Administration: $0{,}60 \times 180 = 108$ paid, 72 owed.
- Cameras: 100 paid, 100 owed until 01/04/2019.
- Loan: interest 72 and principal 300 paid on 31/12/2018; it falls to $1.200 - 300 = 900$ (300 due on 31/12/2019, 600 later).

Step 3. Cash account 2018

Inflows	**m.u.**	**Outflows**	**m.u.**
Collections from customers	2.690	Fertilizers	350
		Hangar rent	300
		Fixed salaries and Social Security	720
		External pilots	192
		Administration (60%)	108
		Thermal cameras (50%)	100
		Loan interest	72
		Loan principal	300
		Dividends	96
Total	**2.690**	**Total**	**2.238**

$$\text{Cash}_{24:00,\,31/12/2018} = 1.955 + 2.690 - 2.238 = 2.407 \text{ m.u.}$$

Step 4. Balance sheet at 24:00 on 31/12/2018

Assets	**m.u.**	**Liabilities and owners equity**	**m.u.**
Land	800	Paid-in capital	4.500
Drones ($2.400 - 375$)	2.025	Losses of 2017 (offset)	0
Software licences ($400 - 100$)	300	Reserves	224
Thermal cameras ($200 - 16$)	184	**Owners equity**	**4.724**
Non-current assets	**3.309**	Bank loan due 2020–2021	600
Stock of fertilizers	100	**Non-current liabilities**	**600**
Accounts receivable	310	Bank loan due 31/12/2019	300
Cash	2.407	Suppliers of fertilizers	150
		Creditors for administration expenses	72

Assets	m.u.	Liabilities and owners equity	m.u.
		Supplier of thermal cameras	100
		Taxes payable	180
Current assets	**2.817**	**Current liabilities**	**802**
Total assets	**6.126**	**Total liabilities and equity**	**6.126**

Accumulated depreciation of the drones: $75 + 300 = 375$.

KEY RESULTS

- Remaining profit 320; dividends 96; reserves 224 m.u.
- Cash 2.407 m.u.
- Owners equity 4.724; current liabilities 802; total 6.126 m.u.

Part (c) · 0.5 points

Compute ROA and ROE of 2018

$$\text{ROA} = \frac{792}{6.126} = 12{,}93\%, \qquad \text{ROE} = \frac{540}{4.724} = 11{,}43\%$$

The ROE is below the ROA because interest and taxes are deducted from the numerator, while the equity base (4.724) is not much smaller than the total assets (6.126).

KEY RESULTS

- ROA = 12,93%; ROE = 11,43% (extension).

Part (d) · 0.75 points

Effect of paying the dividend on 30 June 2019 and of not offsetting the 2017 loss before the 30% payout

Step 1. New distribution

$$\text{Dividend} = 0{,}30 \times 540 = 162, \qquad \text{Reserves} = 540 - 162 = 378 \text{ m.u.}, \qquad \text{the loss of 2017 } (-220) \text{ is not offset}$$

Step 2. Changes in the balance sheet at 31/12/2018

Item	Base case	New case	Change
Cash (the 96 m.u. are no longer paid)	2.407	2.503	+96
Total assets	**6.126**	**6.222**	**+96**
Losses of 2017	0	(220)	-220
Reserves	224	378	+154
Owners equity	**4.724**	**4.658**	**-66**
Dividends payable (current liability)	0	162	+162
Total liabilities and equity	**6.126**	**6.222**	**+96**

Check: $-66 + 162 = +96$ (checked). The shareholders are promised more (162 instead of 96) because the dividend is computed before absorbing the loss; equity is lower by exactly the extra dividend ($162 - 96 = 66$), and the cash stays in the company until June.

KEY RESULTS

- Dividend $0{,}30 \times 540 = 162$ m.u., recorded as "dividends payable" (current).
- Cash +96 (2.503); reserves 378; losses of 2017 (220) remain; owners equity 4.658 (-66).
- Totals 6.222 m.u. (+96).

Interpretation and checks

- **Dimensional check:** farms × m.u./(month·farm) × months gives m.u.; m.u./year × months/12 gives m.u.
- **Accrual vs cash:** net profit 540 m.u., while cash grew by 452 m.u.: $540 + 416$ (depreciation, no cash) -310 (uncollected sales) -100 (stock) $+150 + 72 + 180$ (suppliers, administration and tax still owed) -100 (cameras) -300 (principal) -96 (dividends) $= 452$.
- **Typical mistakes:** expensing all 500 m.u. of fertilizers, amortizing the software in 2017, forgetting the 20% residual value of the cameras, and computing the interest on the original 1.500 m.u.